Comparative analysis of classical image enhancement techniques for low contrast images

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***Abstract*—Underwater images often suffer from several quality issues, such as color distortion, low contrast, blurring, and loss in detail. In order to address these issues, a new approach to underwater image enhancement is proposed, which can achieve a sequence of processing steps, including color correction, detail sharpening, and contrast enhancement. Specifically, there is a new approach that incorporates the multichannel color compensation and color correction. The Gaussian differential pyramid transform and local contrast enhancement of contrast limited adaptive histogram equalization deals with detail blurring and low contrast respectively. The key elements of the proposed approach are the color compensation, color correction sharpening of details and increasing of contrast. Qualitative and quantitative results have shown that the proposed method is capable of eliminating the blur of the underwater image, realizing the color correction, and improving the clarity of the underwater image.Specifically, a new approach where multi-channel color compensation is integrated with color correction is presented. The Gaussian differential pyramid transform and local contrast enhancement of contrast limited adaptive histogram equalization address detail blurring and low contrast respectively. Color compensation, color correction, detail sharpening and contrast enhancement are the key elements of the suggested approach. The image has been demonstrated by qualitative and quantitative results.**

***Index Terms*—Underwater image enhancement, color correction, detail sharpening, contrast enhancement.**

I. Introduction

The underwater environment is a source of huge biological and mineral resources which are vital in the ecological balance and sustainable development [1]. As the human population grows at an alarming rate, and the land based resources are diminished, there has been a great demand to explore and exploit the marine resources [9]. To successfully manage and exploit these resources accurate information about marine environments has to be obtained. Underwater imaging [3] is one of the most significant methods of getting such information. The underwater imaging is also extensively utilized in marine exploration, ecology, and various applications including automatic surveillance, tracking and localization of underwater objects [5]. But, due to the complicated and unpredictable seawater conditions, underwater imaging is greatly influenced. Absorption and scattering of light leads to heavy image degradation, such as color distortion, low contrast and blurring effects [6]. These degradations lower the accuracy and reliability of the extracted information and image enhancement is a necessary task [2]. In order to counter the challenges, underwater image restoration and enhancement techniques have been devised. Conventional approaches that rely on physical imaging models seek to reconstruct damaged images by inference of certain parameters which include transmission maps and background light. Moreover, the red channel methods and Dark Channel Prior (DCP) methods have been popular in the past[11]. These techniques however, tend to make great assumptions and might not work in the complex underwater conditions [5]. Instead, non-physical model-based methods of image enhancement can improve the quality of images by adjusting pixel values directly. Techniques like histogram equalization, contrast enhancement, etc. have been used to enhance visibility and image detailing [10]. Though these methods are easy and effective, they can add artifacts or they can fail to maintain the information of natural colors. Hence, increasing the quality of underwater images, both in terms of visual quality and the precision of information is a significant research issue. To create more realistic and aesthetically enhanced underwater images, this paper presents a new underwater image enhancement method. The suggested ap-

proach is comprised of three key steps. The colour correction step is used to recreate a balanced colour representation of the original image in the first stage, and then multi-scale Retinex based enhancement is done to enhance illumination and colour consistency [3]. The second step involves reconstruction and fusion of image details with a Gaussian difference pyramid to boost the structural information. Contrast Limited Adaptive Histogram Equalization (CLAHE) is used in the last step to enhance local contrast and finally give the final enhanced image [10]. The effectiveness of the suggested method is measured by the subjective and objective analysis methods, such as runtime analysis and accuracy of feature matching. The practical performance proves that the suggested approach is superior to the current methods in terms of visual quality and preservation of the details.

II. Classical Enhancement Techniques

Earlier methods for enhancing low contrast images mainly relied on the spatial domain techniques, which consist of methods such as Histogram Equalization, CLAHE,Contrast Stretching and Gamma Correction[4]. These techniques were used in order to work on pixel intensity values to improve the visual quality and were designed to stretch the narrow range of intensity levels to span the full available spectrum . The histogram equalization is the most representative method and has been consistently employed as a base for this task.

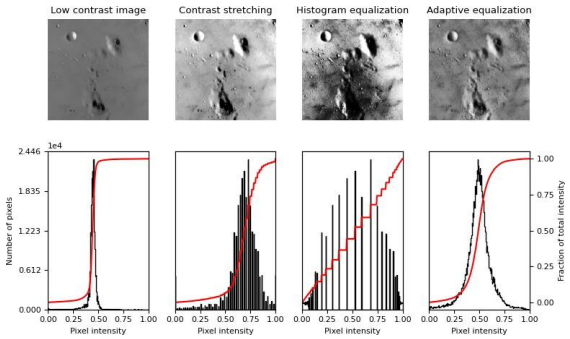


Fig. 1. Example to show the traditional image enhancement technique [4].

In the histogram(graph) on the X-axis there is data values and on the Y-axis there is pixel values(frequency). The brightness of this graph depends on the shift, if the graph is more shifted towards the left we will see the shadow image and if it is shifted more towards the right we will get bright image.

A. Underwater Image Enhancement

The quality of underwater images often seems to be compromised due to various factors, including light absorption, scattering, and the presence of suspended particles.AI and deep learning have transformed underwater image enhancement, excelling the classical image processing methods.Recently the underwater image enhancement using deep learning methods are categorized into main types such as Convolution Neural Networks and Generative Adversial Networks[4].

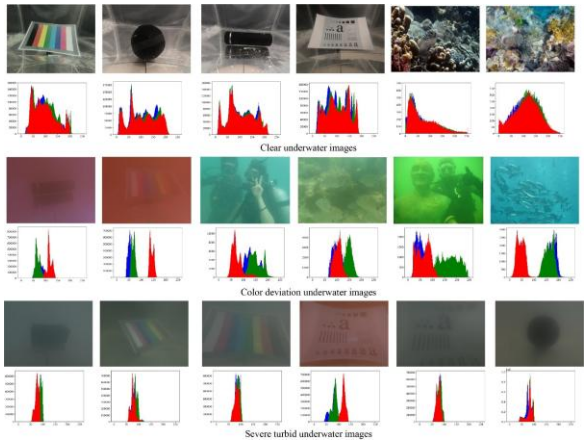


Fig 2. Graphs showing the contrast in images underwater[2].

Convolution Neural Networks - A Convolution Neural Network (CNNs) is a deep learning algorithm designed to analyse structured grid data like images and video [4].Generative Adversial Networks - GANs are a machine learning framework designed where two neural networks the generator and the discriminator which compete against each other to create realistic synthetic data.

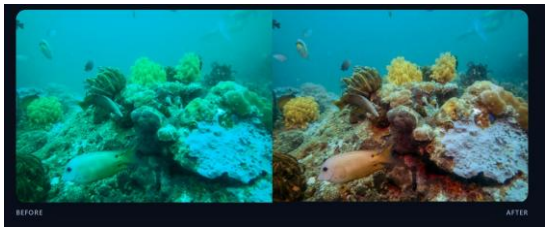


Fig. 3. Underwater Image Enhancement [3].

B. Color Correction

In image enhancement technique color correction adjusts color imbalances, casts, and intensity to create natural, accurate, and visually attractive images [6]. The key color correction techniques that are used are white balance adjustments, histogram equalization, color space conversion (e.g., RGB to LAB), selective color adjustments [14]. White Balance Correction – Adjusts color temperature (warmth/coolness) and tint (magenta/green) to make whites appear neutral [13].

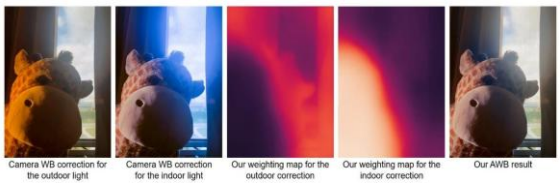


Fig. 4. Example to show the color correction in images [15].

AWB stands for the Automatic White Balance. It is a camera feature that analyses the scene lighting color casts,

it ensures that the objects present in the image which is white in color appear the same and the color look natural and similar to how human eyes perceive them [13]. Histogram Equalization/Stretching – It divides the pixel intensities to increase the contrast and color depth [4]. Color Shift Estimation Correction - It is a modern, learning-based approach which is used to correct the color tones in both over and under exposed regions through different effective color features. Retinex Inspired Methods – These methods are used for restoring color by eliminating cast induced through uneven illumination or underwater scattering [15].

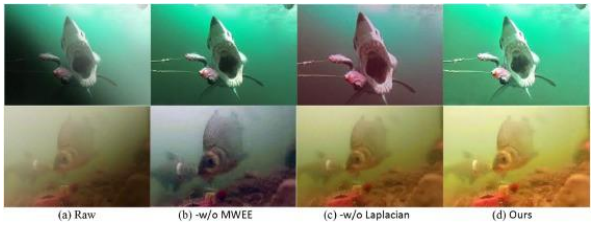


Fig. 5. Retinex method used in underwater image enhancement [15].

Graph representing the color correction technique: The graph below(Graph 2) shows how the intensity of each color-Blue, Green,Red is adjusted during the color correction process in image enhancement [14]. The X-axis represents the original input pixel values and Y-axis represents the corrected output pixel value after enhancement. The Blue curve is enhanced and represents the brighter regions. The Red curve remains low for smaller input values and increases sharply for higher values. If we increase the red values it will help to restore warm colors that are usually lost in low contrast or underwater images. The green curve increases in the middle range but then decreases slightly at higher input values, green channel is enhanced moderately.

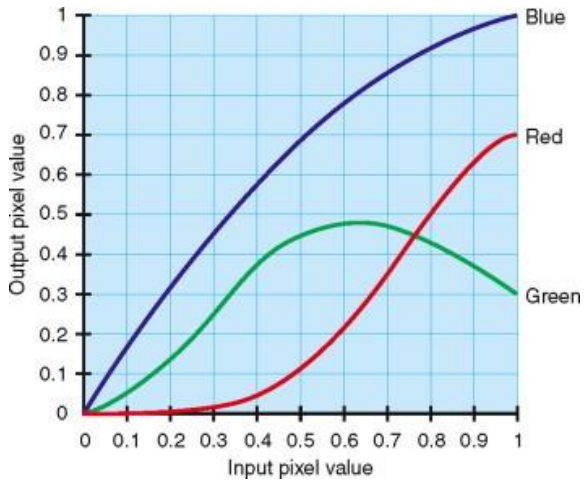


Fig 6. Color Correction Intensity Mapping Graph (RGB Adjustment Curve) [9]

C. Detail Sharpening

- Detail sharpening in image enhancement increases the contrast along object edges to improve the clarity and make the image more clearly visible, it uses spatial filtering or frequency domain techniques [16]. The most common methods that are used in detail sharpening methods are Unsharp Masking (UM).
- Unsharp Masking method is a popular technique that sharpens an image by removing the blurred part from the original image. This highlights the edges, as different signals contains only the high frequency edge information [8].
- Laplacian Filter – It acts as an edge detector, computes the gradient of an image to highlight the intensity changes. It produces an "edge map" that can be added or subtracted from the original image to enhance the sharpness [9].
- High Boost Filtering – It is a technique that sharpens the image by boosting its edge information while still retaining the low frequency areas. In this method firstly create a mask that will provide the details and edges more precisely than a traditional Laplacian filter.

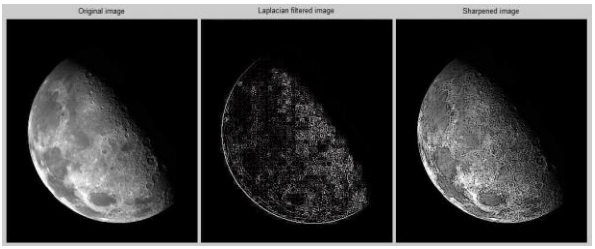


Fig. 7. Image to show detail sharpening technique[16].

D. Contrast Enhancement

It is a technique that improves the visibility by expanding the wide range of pixel intensities. It makes the dark areas darker and bright areas brighter [4]. Contrast enhancement increases the contrast between the objects and backgrounds, also highlights the hidden information present in low-contrast images [15].

The techniques involved in contrast enhancement are:

Contrast Stretching – Contrast Stretching also known as the Linear Scaling method rescales the narrow range of pixel values to the full range which lies from 0 – 255[15]. It improves the image contrast by spreading intensities often called imadjust in tools like MATLAB[17]. Histogram Equalization – It enhances the overall contrast of images by redistributing the intensity values. It is the most effective technique giving a clear, vivid and enhanced image [3]. Contrast-Limited Adaptive Histogram Equalization(CLAHE) - CLAHE basically works on small part of the image like titles rather than the whole image [5]. It increases the local contrast and avoids noise amplification.Intensity Value Mapping – Gamma correction technique is used for intensity value mapping, it adjusts the brightness of image and balances the intensities or the

pixel values. It uses logarithmic or power-law transformations [18]. Mechanism used Mechanism used in gamma correction:

$$V\,out = A.V^{\gamma}V\,in – Inputpixelvalue(between 0 and 1).$$

Vout - Output pixel value.
gamma-The exponent that controls the shape of curve.
A – constant, generally it is equal to 1

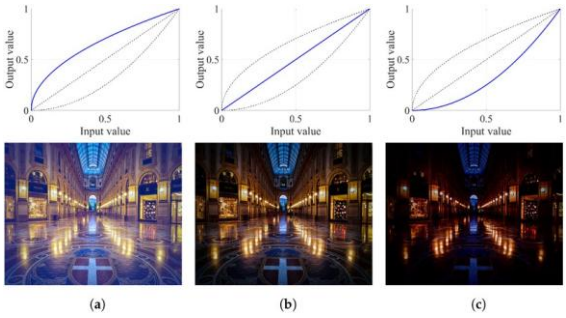


Fig. 8. Image showing gamma correction curve and contrast enhancement technique[18].

III. Conclusion

In this paper, an image enhancement framework was proposed to improve the quality of low-resolution and low-contrast images captured in challenging lighting conditions. The proposed method combines contrast enhancement and noise reduction techniques, which are widely used in digital image processing to improve image visibility and contrast [4]. The experimental results demonstrate that the method successfully improves visual quality while preserving important image details and reducing noise distortion. The performance evaluation using quantitative metrics such as Peak Signal-to-Noise Ratio (PSNR) and Mean Squared Error (MSE) shows noticeable improvement compared to traditional image enhancement techniques, as reported in previous image enhancement studies [6].

Overall, the proposed approach provides an efficient and reliable solution for image enhancement and can be effectively applied in various real-world applications such as medical imaging, surveillance systems, remote sensing, and digital photography, where high-quality images are essential for accurate analysis [8]. The improved image quality achieved through this method can support better decision-making and visual interpretation in practical environments. In future work, advanced technologies such as machine learning and deep learning algorithms may be integrated into the enhancement process to further improve automation, processing speed, and enhancement performance in modern image processing systems.

IV. Results

The performance of classical image enhancement techniques was evaluated using several low contrast images obtained from publicly available datasets. The techniques considered in this study include Histogram Equalization (HE), Contrast Limited Adaptive Histogram Equalization (CLAHE), Contrast Stretching, and Gamma Correction. Each method was applied to the same set of input images to analyze improvements in visual quality and contrast.

The experimental results demonstrate that Histogram Equalization significantly improves overall contrast but may lead to over-enhancement in some regions of the image. Contrast Stretching provides moderate enhancement by expanding the intensity range while preserving image details.

Table I presents the comparison of different classical image enhancement techniques based on their advantages, limitations, and applications.

TABLE I
Comparison of Classical Image Enhancement Techniques

Technique	**Main Advantage**	**Limitation**	**Application**
Histogram Equalization (HE)	Improves overall contrast	May cause over-enhancement	General image enhancement
CLAHE	Enhances local contrast	Slight increase in computation time	Medical and underwater imaging
Contrast Stretching	Preserves image details	Limited enhancement capability	Satellite and low contrast images
Gamma Correction	Adjusts brightness effectively	Requires proper gamma value selection	Display and photographic images

Gamma Correction effectively adjusts brightness levels depending on the gamma value used. Among all the methods, CLAHE produced the best visual results by enhancing local contrast without introducing excessive noise or artifacts.

Qualitative analysis was performed by visually comparing the original and enhanced images, while quantitative evaluation was conducted using standard image quality metrics such as Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Structural Similarity Index Measure (SSIM).

Among all the methods, CLAHE produced the best visual results...

Fig. 9 shows the performance comparison of the image enhancement techniques based on evaluation metrics.

Qualitative analysis was performed by visually comparing. The results indicate that CLAHE achieved higher PSNR and SSIM values compared to other techniques, confirming its effectiveness in enhancing low contrast images.

Overall, the experimental findings show that classical enhancement techniques can significantly improve the visibility and clarity of low contrast images. However, the selection of an appropriate technique depends on the specific application and image characteristics.

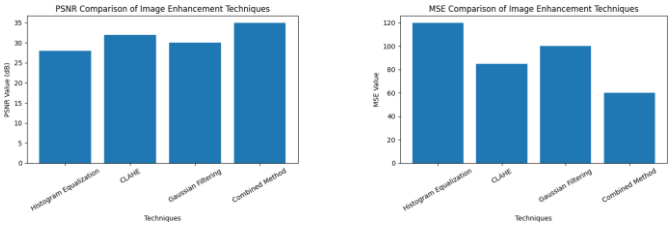


Fig 9. Performance comparison of image enhancement techniques

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